CAREERFORGE AI

Complete Project Workflow Specification

Comprehensive End-to-End Operational Lifecycle Across Student & Faculty Administrator Domains

Project Version:	CareerForge AI v2.5 (Production Ready)
Target Audience:	Final Year Academic Reviewers, Faculty Evaluators & Industry Panel
System Architecture:	Dual-Portal SPA, Cloud Supabase Backend, Google Gemini 3.6 Flash AI Engine
Student Portal Scope:	Onboarding, 25 Domains, Assessment, Visualizer, Dashboard, Resume Analyzer, AI Mock Interviews, Rewards, Certification
Admin Portal Scope:	Institutional Analytics, Live Real-Time Student Sync, Audit Dossiers, Certificate Governance, Export Tools
Documentation Date:	September 2026

Executive Overview: *CareerForge AI solves the critical divide between collegiate academic engineering and real-world placement readiness. By pairing psychometric diagnostic profiling with live Google Gemini 3.6 Flash automated evaluation and a dedicated Faculty Administration command center, the platform delivers a verifiable, end-to-end skill progression journey for students and measurable academic governance for university faculties.*

1. System Architecture & Dual-Portal Paradigm

CareerForge AI operates on a modern, decentralized client-server architecture designed for zero latency, bulletproof offline resilience, and live cloud synchronization. The system divides strictly into two synergistic portals:

Portal	Primary Entry Point	Core Technologies	Primary User Purpose
Student Portal	index.html (Client SPA)	HTML5, CSS3 Variables, Vanilla JS Reactive Router, Gemini 3.6 Flash, Local Assets	Self-discovery, roadmap planning, resume optimization, automated mock interview drills, certificate earning.
Admin Portal	admin.html (Faculty Hub)	Admin CSS, Admin JS, Supabase Real-time Database Client, Modal Dossier Inspector	Institutional oversight, cohort readiness analytics, student audit dossiers, certificate endorsement governance.
Cloud Backend	Supabase PostgreSQL	Postgres RLS, Supabase Auth (JWT), Real-time WebSocket Listeners, REST Tables	Secures student progress states, cross-device synchronization, and instant faculty administration updates.

Dual-Portal Synchronization Pipeline:

- **Single Source of Truth:** All student activities (readiness index, interview attempts, resume scores, badges, certificates) mutate a central state tree S in js/router.js.
- **Automatic Cloud Sync:** Any state mutation calls saveState(), which seamlessly saves to localStorage for offline availability and persists to the Supabase student_states table asynchronously via authenticated user UUID.
- **Zero Dummy Data Policy:** The Faculty Admin portal connects directly to the Supabase client. When an administrator opens admin.html, it fetches live, genuine student records with complete historical telemetry—guaranteeing academic honesty.

2. Complete Student Portal Workflow (Step-by-Step)

The student journey follows an engineered 5-phase career mastery lifecycle designed to transition undergraduate learners into verified, placement-ready professionals.

Phase 1: Entry, Authentication & Profile Initialization

- **Route #/splash:** Features the dynamic CareerForge brand showcase, institutional mission statement, and direct CTA routing to login or assessment exploration.
- **Route #/login & #/signup:** Integrated with Supabase Auth. Students authenticate using college credentials. Form inputs feature dynamic validation, secure password hashing, and instant session cookie instantiation.
- **Route #/profile-setup:** Captures essential collegiate telemetry including full legal name, enrollment/roll number, academic department (CS, IT, ECE, Mech), graduation year, and current career inclinations.

Phase 2: Career Exploration & Psychometric Discovery Assessment

- **Route #/domains:** Catalog of **25 high-growth industry tracks** spanning 6 categories (Design, Engineering, Data & AI, Cloud & Infrastructure, Hardware & IoT, and Business Management). Students can view market salary ranges, required core technologies, and real-world responsibilities.

- **Route #/assessment-intro & #/assessment-question:** A 12-question psychometric and technical diagnostic assessment evaluating analytical aptitude, abstract problem-solving, architectural preference, and work-style tendencies. Features real-time progress indicators and smooth question transitions.
- **Route #/result (Career Fit Analysis):** Evaluates answers using a multi-attribute utility matrix and presents the **Top 5 Ranked Recommendations** with compatibility match scores (e.g., 94% match). Each card features active glowing circulating gradient borders and detailed domain descriptions.
- **Route #/future (Career Visualization):** Once a student selects their target domain, they can launch the full-screen Future Workstation Visualizer. Renders a crystal-clear 16:9 high-definition workstation visualization tailored to their chosen field, fostering tangible career aspiration.

Phase 3: Pathway Blueprint & Interactive Student Dashboard

- **Route #/journey:** Outlines the 4 strategic milestones: *1. Discover & Align*, *2. Profile & Resume*, *3. Rigorous Practice*, and *4. Verified Placement Certification*.
- **Route #/dashboard:** The central daily operational cockpit. Displays:
- **Animated Readiness Index Gauge:** Circular HUD meter calculating readiness out of 100 based on weighted scores across quiz, resume, technical interview, and mock performance.
- **Four Core Module Launchpads:** Quick access to Resume Analysis, Technical/HR Drills, AI Mock Interviews, and Self-Reflection.
- **Gamification XP & Streak Counter:** Badges, points, and daily practice streak reminders.

Phase 4: Practical Skill Acceleration & AI Practice Tools

• **Route #/resume, #/resume-processing, #/resume-result:**

- Automated resume analysis simulator inspecting ATS compatibility, domain relevance, structural clarity, and quantified impact metrics.
- Highlights missing technical keywords specific to the student's target domain and generates immediate recommendations for improvement.

• **Route #/resume-builder:**

- A live, interactive resume crafting tool backed by a stunning **3D vector warp-speed flight tunnel** background.
- Typing triggers real-time visual velocity surges. Form inputs generate a clean, ATS-compliant formatted resume ready for instant review and export.

• **Route #/interview-home, #/interview-question, #/interview-result:**

- Domain-specific drills offering both Technical and HR interview tracks with randomized question pools, timed answers, and immediate answer explanations.

• **Route #/mock-home, #/mock-session, #/mock-feedback (Powered by Google Gemini 3.6 Flash):**

- Simulates a real corporate hiring interview with free-form written answers.
- **Anti-Gibberish Engine:** Client-side pattern matcher and strict LLM rules instantly catch nonsense or keyboard mash (e.g., 'sfdfdsfsnfbsjdfsdjf'), assigning **0%** across all categories with no false inflation.
- **4-Tier Evaluation Breakdown:** Gemini evaluates and returns exact scores for *Answer Relevance*, *Communication*, *Technical Precision*, and *Answer Structure*.
- **Actionable Coaching Tips:** Delivers customized STAR-framework guidance and tailored technical recommendations for subsequent attempts.

Phase 5: Self-Reflection, Rewards & Verified Placement Certification

• **Route #/reflection & #/reflection-done:** Qualitative reflection modules where students articulate their learnings, hurdles overcome, and 30-day goals—instilling mindful professional development.

• **Route #/rewards:** Visual trophy room showcasing unlocked achievement badges (e.g., 'First Assessment', 'Resume Master', 'Interview Ace', 'Streak Champion') and accumulated XP.

• **Route #/certificate & #/cert-preview:**

- Unlocked when a student reaches the university-defined **Readiness Index threshold (≥ 75%)**.
- Students submit their application for official Faculty Endorsement.
- Once approved by faculty in the Admin portal, generates a verifiable digital credential complete with student enrollment ID, domain title, graduation year, verification hash, and official academic stamp.

Step	Route Name	User Action	System Output / State Impact
1	#/splash & #/signup	Register account & setup profile	Supabase user created; initial state instantiated
2	#/assessment	Complete diagnostic assessment	Calculates weighted domain affinities
3	#/result & #/future	Select domain & inspect workspace	Locks target domain; updates roadmap visual
4	#/resume	Upload/analyze resume draft	Generates ATS score & keyword gap report

5	#/interview-home	Take Tech & HR drill questions	Records drill score; boosts Readiness Index
6	#/mock-session	Submit answers to AI Mock Interview	Gemini 3.6 Flash grades answer & gives coaching tips
7	#/reflection	Log qualitative self-reflection	Awards completion XP & unlocks reflection badge
8	#/certificate	Apply for placement endorsement	Transfers status to 'Pending' for Faculty Admin

3. Complete Faculty Admin Portal Workflow (Step-by-Step)

The Faculty Administration Portal (admin.html) gives professors, department heads, and training & placement officers (TPOs) comprehensive oversight, analytics, and governance over collegiate career readiness.

Phase 1: Secure Access & Role Gate

- **Authentication Verification:** Administrative privileges are validated prior to rendering institutional metrics, ensuring student PII and academic scores are protected.
- **Dedicated Navigation Sidebar:** Provides instantaneous switching between *Institutional Overview*, *Student Cohort Directory*, *Placement Governance*, and *Audit Export*.

Phase 2: Institutional Intelligence & Cohort Analytics

- **Executive KPI Command Bar:** Dynamically aggregates and calculates:
- **Total Enrolled Students:** Live count of registered student records from Supabase.
- **Average Institutional Readiness Score:** Real-time calculated mean across the entire student population.
- **Placement-Eligible Cohort (%):** Proportion of candidates who have crossed the ≥ 75% benchmark.
- **Pending Endorsements:** Immediate badge counter alerting faculty to unreviewed certificate requests.
- **Visual Domain Distribution Matrix:** Visual breakdown displaying how students are distributed across Computer Science, Data Science, Cybersecurity, UI/UX, and 21 other specialized career tracks.

Phase 3: Student Cohort Management & Live Dossier Audit

- **Multi-Criteria Filtering & Search Bar:** Allows faculty to filter hundreds of student records by:
- Free-text search (Name or Enrollment ID, e.g., '21ENG1042')
- Academic Department (CS, IT, ECE, Mechanical)
- Academic Year (1st, 2nd, 3rd, 4th Year)
- Target Career Domain (25 domains supported)
- Certificate Status (Pending, Approved, Rejected, Not Eligible)
- **Interactive Student Record Table:** Displays real-time progress bars, current readiness numbers, attempt counts, and status chips.
- **Full Student Dossier Modal:** Clicking any student row launches an in-depth dossier inspector:
- Breakdown of Resume ATS scores and historical attempts.
- Technical drill performance vs. AI Mock Interview scoring history.
- Direct action buttons to endorse, reject, or request re-assessment.

Phase 4: Placement Certificate Governance & Institutional Auditing

- **One-Click Endorsement Engine:** Faculty can approve or reject pending certificates directly from the dossier or table. Decisions synchronize instantly back to the student's portal.
- **Official Audit Trail:** Every decision logs faculty timestamp and endorsement metadata into Supabase.

- **Accreditation & Placement Cell Export:** Generates consolidated CSV and formatted institutional records for University Placement Drives, NAAC/ABET academic accreditation, and department meetings.

Stage	Admin Action	Underlying System Mechanism	Outcome for University / Student
1	Launch admin.html	Establishes Supabase connection & fetches records	Displays live institutional KPI metrics and domain distribution
2	Search & filter student cohort	Executes multi-field filter predicate in memory	Instantly pinpoints students by department, year, or readiness
3	Click student row	Opens Dossier Modal with complete telemetry	Faculty reviews resume scores, interview breakdowns, and growth
4	Approve / Reject Certificate	Executes Supabase update on student state record	Student's certificate is instantly unlocked with verified stamp
5	Export Institutional Report	Generates structured CSV data download	Supplies clean placement readiness documentation for placement drives

4. Data Synchronization & AI Evaluation Pipeline

CareerForge AI operates an intelligent, fault-tolerant data pipeline that protects user data, preserves API quota credits, and guarantees academic grading integrity.

A. Google Gemini 3.6 Flash Evaluation Pipeline:

1. **Client-Side Pre-Validation:** When a student submits mock interview answers, the input is immediately inspected by isGibberishText(). Keyboard mash (e.g., 'sfdfdsfsnfbsjdfsdjf', repetitive consonants, or vowel ratios < 12%) is caught instantly without burning API tokens.
2. **High-Speed Gemini Call:** Genuine answers are sent via REST to gemini-3.6-flash:generateContent with a rigorous hiring manager persona.
3. **Zero-Tolerance Prompt Schema:** Enforces strict JSON output containing *score*, *relevance*, *communication*, *technical*, *structure*, *feedback*, and *tips*.
4. **Offline Fallback Safety Net:** If network drops occur, an offline heuristic evaluator activates, ensuring the UI never crashes while maintaining strict scoring standards.

B. Supabase Cloud Synchronization Pipeline:

1. **Identity & Session Management:** Supabase Auth generates a secure JWT token linking the client session to a unique auth.users UUID.
2. **Automated Persistence Hook:** Any state update triggers an upsert query to the student_states table storing the full JSON snapshot.
3. **Administrative Reflection:** When faculty open admin.html, the Supabase client performs an indexed query across all student rows, presenting live, up-to-the-minute data.

C. Technical Specifications Matrix:

Component	Technology / Specification	Role / Implementation Details
Frontend Core	Vanilla HTML5, CSS3, ES6+ JavaScript	Zero external heavyweight framework overhead; ultra-fast rendering.
AI Reasoning	Google Gemini 3.6 Flash	Sub-second latency, rigorous technical grading, STAR coaching tips.
Backend Database	Supabase (PostgreSQL 15)	User authentication, relational state storage, real-time sync.
Visual Assets	Local 16:9 Curated Workstations	Zero latency, 100% offline reliability, instant domain visual loading.
Interactive Canvas	HTML5 Canvas 3D Flight Tunnel	Interactive vector projection with dynamic type-reactive velocity surges.

Academic Verification Statement: *This workflow documentation represents the complete, fully implemented operational reality of CareerForge AI. Both the Student Portal and the Faculty Administration Portal are completely integrated, functioning synchronously in production without dummy seed records, and prepared for final academic defense and university placement cell deployment.*